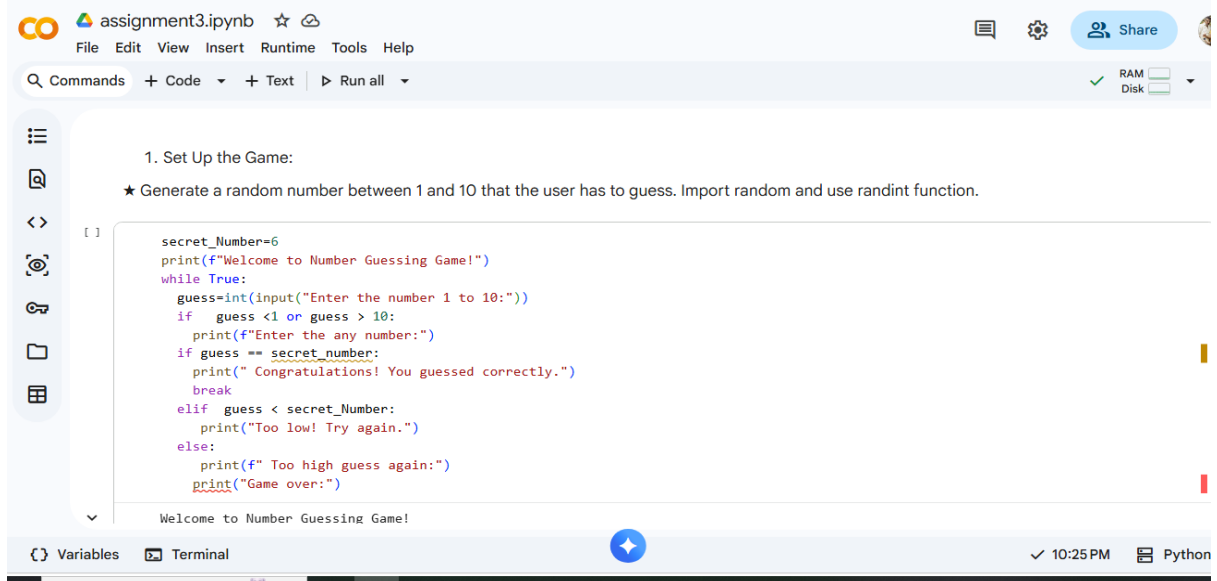


PYTHON ASSIGNMENT-3

USING FOR LOOP,WHILE LOOP, AND FUNCTION

1.SET UP GAMES



The screenshot shows a Jupyter Notebook titled "assignment3.ipynb". The interface includes a top bar with "File", "Edit", "View", "Insert", "Runtime", "Tools", and "Help" menus. Below the menu bar is a "Commands" search bar and a "Run all" button. On the right, there are icons for chat, settings, and a "Share" button. The left sidebar contains icons for file explorer, search, and other notebook functions. The main area displays a code cell with the following Python code:

```
[ ] secret_Number=6
print(f"Welcome to Number Guessing Game!")
while True:
    guess=int(input("Enter the number 1 to 10:"))
    if guess <1 or guess > 10:
        print(f"Enter the any number:")
    if guess == secret_number:
        print(" Congratulations! You guessed correctly.")
        break
    elif guess < secret_Number:
        print("Too low! Try again.")
    else:
        print(f" Too high guess again:")
    print("Game over:")
```

Below the code cell, the output "Welcome to Number Guessing Game!" is visible. The bottom of the interface shows a "Variables" panel, a "Terminal" panel, and a status bar indicating the time as 10:25 PM and the language as Python.

PROMPT THE USER

The first screenshot shows a Jupyter Notebook titled 'assignment3.ipynb'. The code cell contains the following Python code:

```
secret_number = 7
attempts = 3

print("Welcome to Number Guessing Game!")

while attempts > 0:
    guess = int(input("Guess a number (1 to 10): "))
```

The output cell shows the following text:

```
Welcome to Number Guessing Game!
Enter the number 1 to 10:6
Too high guess again:
Game over:
Enter the number 1 to 10:7
Congratulations! You guessed correctly.
```

The second screenshot shows the same Jupyter Notebook with the code cell updated to include an if statement to check if the guess is correct and update the attempts counter:

```
if guess == secret_number:
    print("Congratulations! You guessed correctly.")
    break
else:
    attempts -= 1
    print("Wrong guess! Try again.")
    print("Attempts remaining:", attempts)

if attempts == 0:
    print("Game Over! The number was", secret_number)
    break
continue
```

The output cell shows the following text:

```
Welcome to Number Guessing Game!
Guess a number (1 to 10): 3
Wrong guess! Try again.
Attempts remaining: 2
Guess a number (1 to 10): 5
Wrong guess! Try again.
Attempts remaining: 1
...
```

CONTROL STATMENT

The screenshot shows a Jupyter Notebook titled 'assignment3.ipynb'. The code cell contains the following Python code:

```
secret_number = 7
attempts = 3

print(" Welcome to Number Guessing Game!")

while attempts > 0:
    guess = int(input("Guess a number (1 to 10): "))

    if guess < 1 or guess > 10:
```

The output cell shows the following text:

```
Attempts remaining: 2
Guess a number (1 to 10): 5
... Wrong guess! Try again.
Attempts remaining: 1
Guess a number (1 to 10): 8
Wrong guess! Try again.
Attempts remaining: 0
Game Over! The number was 7
```

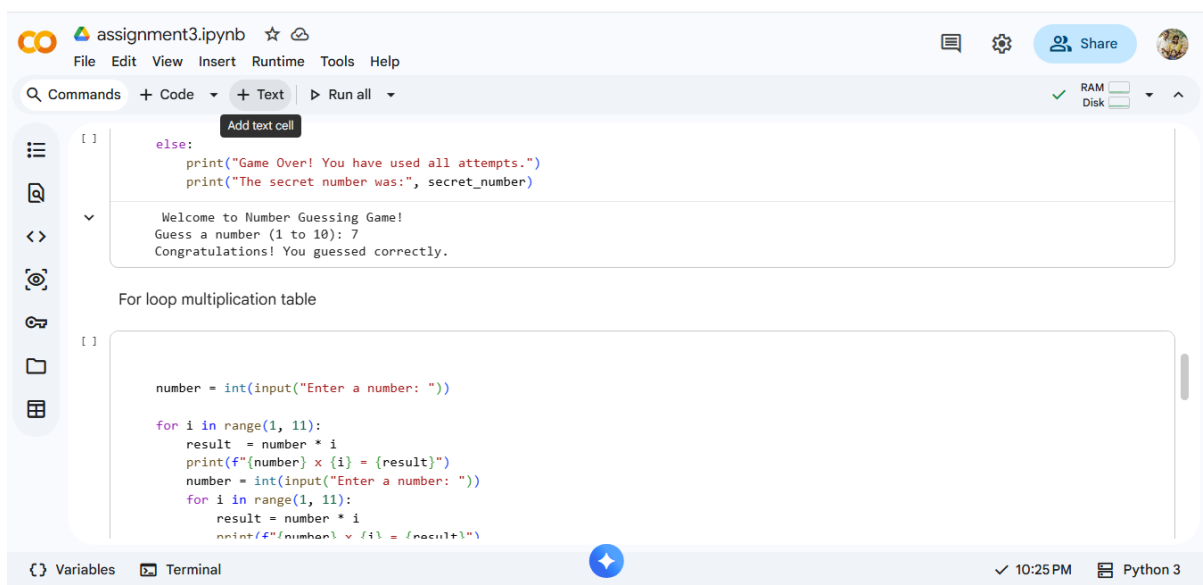
The text 'Control statment' is visible in the output cell.



assignment3.ipynb

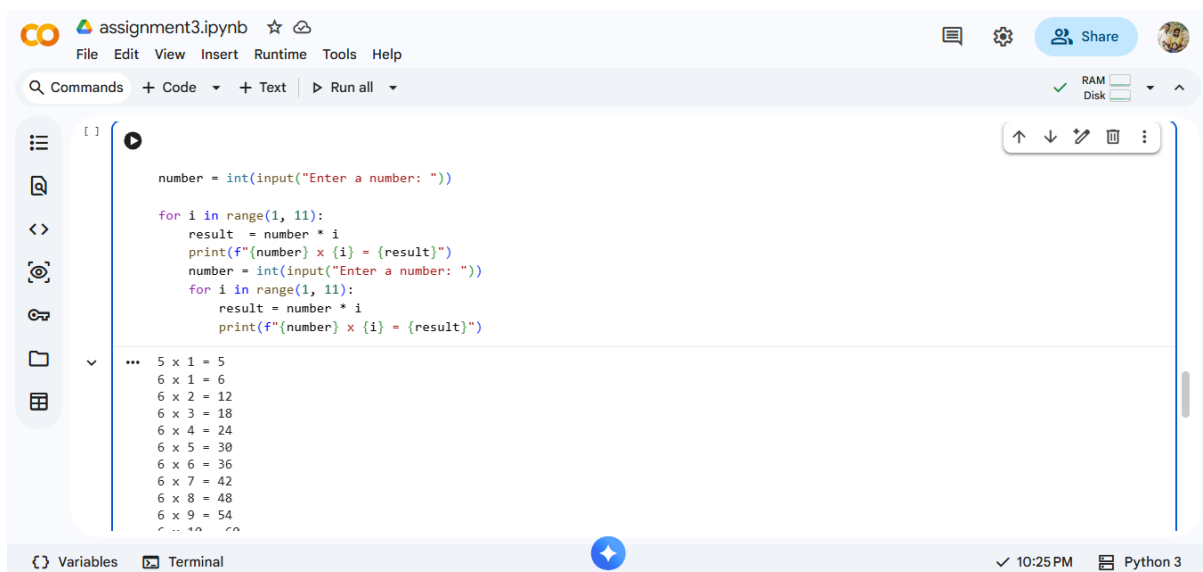
```
[ ] |  
    continue  
  
    if guess > secret_number:  
        print("Too high! Try again.")  
  
    elif guess < secret_number:  
        print("Too low! Try again.")  
  
    else:  
        print("Congratulations! You guessed correctly.")  
        break  
  
    attempts -= 1  
    print("Attempts remaining:", attempts)  
  
    else:  
        print("Game Over! You have used all attempts.")  
        print("The secret number was:", secret_number)  
  
Welcome to Number Guessing Game!  
Guess a number (1 to 10): 7  
Congratulations! You guessed correctly.
```

FOR LOOP MULTIPLICATION



assignment3.ipynb

```
[ ] |  
    else:  
        print("Game Over! You have used all attempts.")  
        print("The secret number was:", secret_number)  
  
Welcome to Number Guessing Game!  
Guess a number (1 to 10): 7  
Congratulations! You guessed correctly.  
  
For loop multiplication table  
  
[ ] |  
  
    number = int(input("Enter a number: "))  
  
    for i in range(1, 11):  
        result = number * i  
        print(f"{number} x {i} = {result}")  
        number = int(input("Enter a number: "))  
        for i in range(1, 11):  
            result = number * i  
            print(f"{number} x {i} = {result}")
```



assignment3.ipynb

```
[ ] |  
    number = int(input("Enter a number: "))  
  
    for i in range(1, 11):  
        result = number * i  
        print(f"{number} x {i} = {result}")  
        number = int(input("Enter a number: "))  
        for i in range(1, 11):  
            result = number * i  
            print(f"{number} x {i} = {result}")  
  
... 5 x 1 = 5  
6 x 1 = 6  
6 x 2 = 12  
6 x 3 = 18  
6 x 4 = 24  
6 x 5 = 30  
6 x 6 = 36  
6 x 7 = 42  
6 x 8 = 48  
6 x 9 = 54  
6 x 10 = 60
```

CALCULATE WEIGHT AND HEIGHT

```
[1]
✓ 34s
def calculate_bmi(weight, height):
    bmi = weight / (height ** 2)
    return bmi

weight = float(input("Enter your weight in kg: "))
height = float(input("Enter your height in meters: "))
bmi = calculate_bmi(weight, height)
print(f"Your BMI is: {bmi:.2f}")

def calculate_bmi(weight, height):
    bmi = weight / (height ** 2)
    return bmi

weight = float(input("Enter your weight in kg: "))
height = float(input("Enter your height in meters: "))
bmi = calculate_bmi(weight, height)

▼ ... Enter your weight in kg: 60
Enter your height in meters: 13.5
Your BMI is: 0.33
Enter your height in meters: 1.6
```